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Measuring the Human Dimensions of Digital Leadership for Successful Digital Transformation

Digital leaders can use the authors' Digital Leadership Scale to assess their own readiness and ability to accelerate digital transformation.

Haroon Abbu, Paul Mugge, Gerhard Gudergan, Gerrit Hoeborn, and Alexander Kwiatkowski

OVERVIEW: The digital transformation of organizations continues at a frenetic pace. While some companies have achieved trailblazer status, others are lagging because they find the transition difficult. Digital leaders play a pivotal role in digital transformation successes because they can inspire organizational and employee confidence in these disruptive and sometimes risky initiatives. We present a Digital Leadership Scale with 15 dimensions. Our scale focuses on successful leadership in digital transformation. We aimed to capture the qualities of leaders leading digitally mature organizations so that leaders of digitally developing organizations can increase their likelihood of success. We also mention the future development of a Digital Leadership Scale self-assessment tool that digital leaders can use to assess their own readiness and ability to accelerate digital initiatives.

KEYWORDS: Digital leadership, Trust, Leadership assessment, Digital transformation

Digital transformation is more about people than it is about digital technology. It requires organizational changes that are backed by leadership and driven by radical challenges to corporate culture. Organizations undergoing digital transformation fall into two categories: *digitally developing* and *digitally mature* (Gudergan et al. 2019; Mugge et al. 2020). Digitally mature organizations are managed differently than digitally developing organizations. Their leaders align human and financial resources with the company's strategy; create a collaborative, and nimble development environment; and promote open and transparent communication (Mugge et al.

2020). Digital transformation requires leaders to build trust (Mugge, Abbu, and Gudergan 2021; Gudergan et al. 2021; Abbu et al. 2020). The attributes digital leaders exhibit are the main difference between digitally mature and digitally developing companies.

Digital leadership is multidimensional (Prince 2018); it comprises elements of authentic leadership, transactional leadership, and transformational leadership. In their literature review of digital leadership, Eberl and Drews (2021) argue that a practice-oriented and integrated scale is needed to measure digital leadership. Currently, no established

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practical method exists to assess the human attributes of successful digital leaders. Our aim is to develop a Digital Leadership Scale, a practical tool to assess digital leaders’ human dimensions.

Study

We conducted a study to ascertain which human qualities influence digital leaders’ performance. We interviewed 13 successful leaders of digitally mature organizations. We consider them successful based on their track record of successful digital transformations in their companies, as evidenced in published case studies and other articles. We wanted to hear directly from them regarding what they are doing and what advice they have for other digital leaders. Our goal is to take the learnings from leaders of digitally mature organizations and help leaders of digitally developing organizations progress toward digital maturity.

Method

To find suitable interviewees, we used the maximum variation strategy of purposeful sampling (Palinkas et al. 2015). The variation factors were geographic representation, size of the company, gender, and industry representation. We developed a database of the leaders we wanted to interview. Our list comprised leaders associated with companies of any size, from a micro-startup to a conglomerate, in any industry, headquartered in either the United States or Germany. We made a deliberate attempt to include female digital leaders to balance the perspectives and to understand if it is harder for them to build trust in organizations where men hold most leadership positions.

Between August and October 2020, we conducted 13 semi-structured interviews (3 women, 10 men). At least two members of the research team participated in each interview. We conducted interviews face-to-face and via Zoom. We asked interviewees five questions that we felt would identify what actions leaders are taking and how and why they pursue them:

- 1. What is your key theme (main message) for transforming the organization?
- 2. What is your organization’s digital strategy and vision?
- 3. What actions have you taken to achieve your theme?
- 4. What results have you witnessed to date?
- 5. What advice do you have for other digital leaders?

We audio-recorded the interview with the interviewees’ consent and transcribed them for data analysis. Participation was voluntary and without compensation. The interviewees hold various senior management

positions: seven CEOs, two business unit heads (BUHs) developing digital products, two functional heads (FHs) leading corporate functions, and two chief digital officers (CDOs) (Table 1). Together, the digital leaders we interviewed represented companies with combined revenues of more than \$225B and nearly 1.2 million employees.

We used an inductive approach to reduce the extensive text data gained from the interviews (Thomas 2006). We reviewed the interview transcripts several times to identify and code the categories. To identify leadership traits specific to digital transformation, we used conventional content analysis, which is exploratory and allows the categories and names for categories to flow from the data (Hsieh and Shannon 2005).

We based our interpretations solely on what the interviewees said. We reviewed the transcripts for direct quotations, analyzed relevant text, and reviewed the final interview report with the interviewees to confirm each quotation’s accuracy. Using content analysis—both automated and manual—we conceptionally analyzed the presence of certain words, themes, or concepts within the text of our interviews and then summarized them into manageable code categories. We first coded the text automatically using software to identify possible categories. We then took the terms that fit the dimensions and manually searched the interview text to see if the interviewee mentioned them or implied them. We counted the number of participants mentioning each dimension and created a table of our content analysis results (Table 2).

Results

To provide context regarding human dimensions of digital leadership, we highlight a few quotations from our participants. We highlight keywords in italics for illustrative purposes.

The managing director of a management consulting company said, “The most successful leaders that I’ve seen have an innate ability to connect the dots—to be systems thinkers—and can *communicate* extremely effectively to their team

TABLE 1. Interviewee positions and industry representation

Industry	Revenue US\$	Employees	Position
Agriculture	< 50 million	< 250	CEO
Automotive	> 25 billion	> 35,000	Functional Head (FH)
Consulting	> 40 billion	> 500,000	Business Unit Head (BUH)
Energy	< 50 million	< 250	CEO
IT Consulting	> 75 billion	> 350,000	CDO
Luxury Goods	> 50 billion	> 160,000	CEO
Manufacturing	> 500 million	> 500	CEO
Market Research	< 50 million	< 250	CEO
Mechanical Engineering	< 50 million	< 250	CEO
Medical technology	> 10 billion	> 50,000	Functional Head (FH)
Pharmaceutical	> 25 billion	> 25,000	CDO
Business Services	> 1 billion	> 50,000	CEO
Software	> 1 billion	> 5,000	Business Unit Head (BUH)

TABLE 2. Results of content analysis

#	Dimension	Search Terms	Mentioned By
1	Honesty	Honesty, integrity, openness	7
2	Humility	Humility, not self-oriented, accept/give feedback, give me advice, respect for people	6
3	Courage	Courage, deal with blind spots, willing to change, embracing change, freedom of action	7
4	Ethical AI	Ethics in AI, fairness, explainability, mitigate bias	4
5	Growth Mindset	Mindset	7
6	Transparent Agenda	Agenda, transparent, openness, clear, clarity	7
7	Data Focus	Data, analytics	7
8	Inspire Engagement	Engage, empower, motivate, get buy-in, desire	7
9	Storytelling	Storytelling, visual analytics, talk-about, celebrate wins	5
10	Digital Literacy	Digital technology, connecting to digital, digital capabilities, technology company, know-how	7
11	Positive Attitude	Positive attitude	5
12	Skills Acquisition	Skills, digital skills, address skill gaps	8
13	Knowledge Sharing	Knowledge sharing, knowledge	9
14	Participative Style	Style, servant leadership, team, people-first, put yourself in their shoes, people respect what you inspect	8
15	Track Record	Track record, outcomes, results	13

and to their clients. A digital leader thus also needs to be a good *storyteller*.” The CEO and founder of a German, family-owned mechanical engineering business suggests, “Be genuinely interested in your employees. I expect absolute *transparency* and *openness*. I encourage employees to speak their minds and share their opinions. I say what fits and what doesn’t fit. And I expect the same from my employees.” A technology company CTO noted, “As we consider putting the AI to work, we have a responsibility to ensure *fairness* and to make sure that we can explain how the model arrives at a recommendation.”

The CEO of an energy company said, “Trust has created a much greater willingness to provide relevant *data* and *information* to each other. This type of information-sharing can drive product optimizations with tangible customer benefit.” The director of strategy for an automotive company said, “You need role models and people who have actual experience in digital. In the end, it is essential for digital leaders not just to talk about digitalization, but also to prove they are able to *deliver results*.”

Digital Leadership Scale

Based on our results, we developed a Digital Leadership Scale consisting of 15 human dimensions. For each dimension, we provide a definition based on relevant literature, discuss prior research pertaining to its relevance to digital leadership, and list existing scales, if any. We provide a practical definition to ensure it is applicable for digital leadership. We then list three questions (“items”) digital leaders can use to measure each dimension. We considered items from existing scales and constructs only conceptually while creating new items for each of the 15 dimensions, since it is not good practice to detach items from their original scale context (Kraus and

Kreitenweis 2020). Consistent with our prior research, we have categorized these 15 dimensions according to Stephen Covey’s (2006) 4 Cores of Credibility: Integrity, Intent, Capabilities, and Results (Figure 1).

#1: Honesty

Rotter (1967, p. 651) defines honesty as “the expectancy that the word, promise, verbal or written statement of another individual or group can be relied upon.” Servant leadership, for example, values honesty highly (Russell and Stone 2002). A major study by IBM showed that role modeling by executives is the most important factor for successful change initiatives (Jorgensen, Bruehl, and Franke 2014), and role models must be credible and embody what they say. Honesty is the most important factor to generate credibility in a leader (Kouzes and Posner 2011). The measurement of honesty is part of the HEXAGO Personality Scale (Ashton and Lee 2009) and Satow’s B5T scale (Satow 2012). For measuring honesty in the context of digital transformation, we are concerned with how it manifests interpersonally, especially how the digital leader conveys honesty and how it contributes to digital transformation success.

Working Definition: An assessment of the leader’s moral character that connotes positive and virtuous attributes such as integrity, truthfulness, and straightforwardness. Honesty also involves being trustworthy, loyal, fair, and sincere.

As a leader of digital initiatives, I ...

1. Address concerns employees have with the digital strategy publicly.
2. Give employees clear answers, even to critical questions.

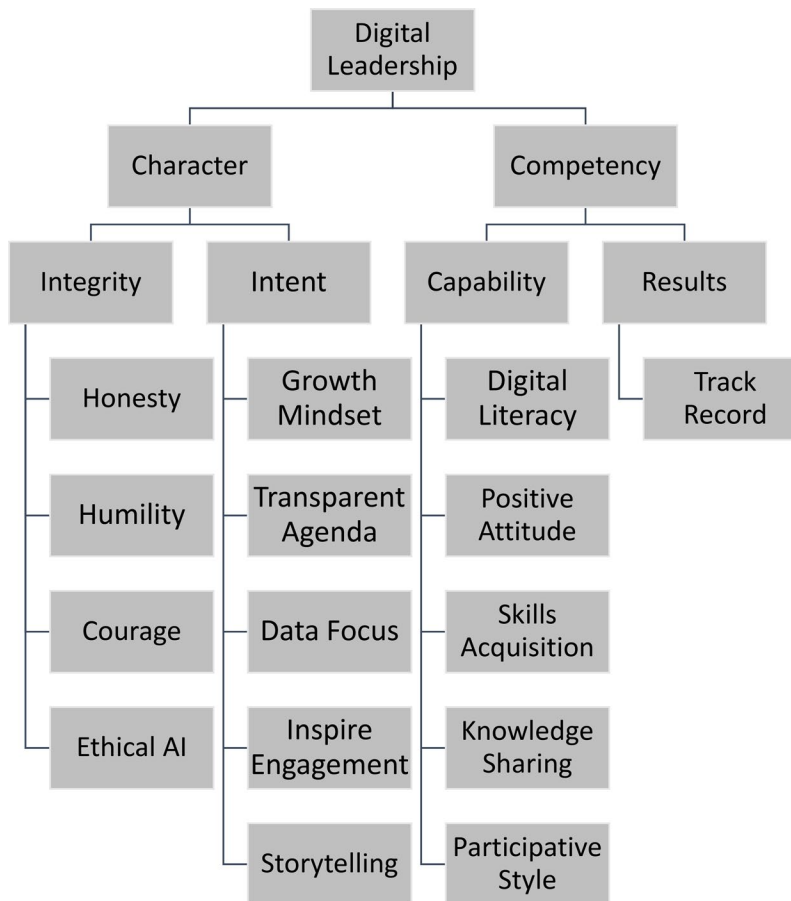


FIGURE 1. Framework and dimensions of the proposed Digital Leadership Scale (adapted from Covey [2006])

3. Do my best to talk straight, and I expect the same from my employees.

#2: Humility

From an interpersonal perspective, humility “involves regulation of egotism and cultivation of other-orientation rather than a selfish preoccupation with one’s own needs” (Davis et al. 2017, p. 4). Leaders manage to build trust through selfless behavior. According to the servant leadership theory, a good leader’s most important task is to serve other people (Baggett 1997). Humility is a necessary dimension to understanding employees’ true needs first. This dimension does not negate leaders representing their

own interests; however, humility implies putting aside one’s own opinion and listening empathetically to employees to gain new insights, even if differences of opinion exist. Neilsen and Marrone (2018) have studied the components included in the humility concept. Humility is also part of the HEXAGO Personality Scale (Ashton and Lee 2009). In the context of digital transformation, we are focused on how the leader displays humility externally within a business interaction.

Working Definition: An assessment of the leader’s modest view of their own importance—that is, their humbleness.

As a leader of digital initiatives, I ...

1. Respect the opinions of our employees, even if it is not the same as mine.
2. Value every contribution of every individual, no matter how big or small it may be.
3. Gratefully accept any feedback both positive and negative.

#3: Courage

Woodard (2004, p. 174) defines courage as “the ability to act for a meaningful (noble, good, or practical) cause, despite experiencing the fear associated with perceived threat

In the context of digital transformation, we are focused on how the leader displays humility externally within a business interaction.

exceeding the available resources.” Change requires courage because it may break up deeper and more extensive entrenched structures. Sanders (1994) said leaders need the courage of the highest order. An anxious leader tends to ignore the threatening aspects of changing circumstances, which may, in turn, trigger a chain reaction among employees throughout the company. Courage is also a prerequisite for successfully mastering the dimension of honesty. Employees want leaders who have the courage to be honest, especially in difficult circumstances, even if that honesty could prove detrimental to the leader (Russell and Stone 2002). Comprehensive measurement constructs exist for assessing courage (for example, Woodard and Pury’s [2007] The Construct of Courage scale).

Working Definition: An assessment of the leader’s mental or moral strength to venture, persevere, and withstand danger, fear, or difficulty.

As a leader of digital initiatives, I ...

1. Change the plan when presented with new, disruptive market realities.
2. Do the right thing for our company even if this could have unpleasant consequences.
3. Pursue the strategy even when I know it will be difficult to achieve.

#4: Ethical AI

Recent advances in machine learning and neural networks have enabled artificial intelligence (AI) to be used for hiring, criminal justice, health care, and other applications. The prevalence of AI in many industries has brought AI ethics to the forefront of digital leadership. As promising as machine learning technology is, DeBrusk (2018) points out how it can also be susceptible to unintended biases that require careful planning to avoid. Ethical leadership represents shared facets with authentic, spiritual, and transformational leadership such as altruism, integrity, and role modeling (Brown and Treviño 2006). Existing scales that measure ethics, such as the Ethical Leadership Scale (Brown, Treviño, and Harrison 2005) and the Leadership Virtues Questionnaire (Riggio et al. 2010), lack an ethical technology assessment. Ethical concerns related to the application of AI in data usage and managerial activities is an important topic for digital leaders to address (Cortellazzo, Bruni, and Zampieri 2019).

Working Definition: An assessment of a leader’s ability to do everything in their power to ensure the results of AI and other technologies employing machine learning models, are fair and explainable.

As a leader of digital initiatives, I ...

1. Make sure senior management understands the need for building trustworthy AI that ensures fairness and accuracy, which meets their privacy and compliance requirements.
2. Put processes in place to ensure no human biases creep into algorithms and affect outcomes.
3. Develop and encourage participation in training sessions related to ethics in AI.

#5: Growth Mindset

A growth mindset explores the extent to which people believe they can become smarter if they work at it. It is based on the belief that “your basic qualities are things you can cultivate through your efforts” (Dweck 2016, p. 7). Research shows that individuals with a growth mindset develop important skills for managing change in a professional context, such as more effective problem-solving strategies and dealing with feedback (Gottfredson and Reina 2020). In the area of digital transformation, leaders with a growth mindset look for new opportunities to realize their visions (Carpenter 2012), which requires that they focus on creative, innovative, and sometimes risky tactics (Renko et al. 2012). A growth mindset is also important to motivate one’s employees. For example, Petrucci and Rivera (2018) and Prince (2018) emphasize the relevance of a growth mindset for digital leaders in that it encourages employees to be curious, think alternatively, and expand their knowledge in accordance with “lifelong learning.” Dweck’s (1999, 2016) Growth Mindset Scale consists of items that focus on the belief that intelligence has a fluid component, that it can grow. We have adopted this basic idea and changed it contextually for a business environment.

Working Definition: An assessment of the leader’s desire to tackle new challenges and grow as a person.

As a leader of digital initiatives, I ...

1. Don’t view failure as a disappointment but as a learning experience that can lead to change.
2. Believe that all people, including myself, can develop their capabilities to do new things, regardless of where they come from.
3. Accept new challenges, take advantage of feedback, and provide timely feedback to subordinates.

#6: Transparent Agenda

Transparency includes openness, freedom of information, clarity, accuracy, and the timely release of relevant information (Bernstein 2017). It enables employee participation and informs them of the opportunities and risks of digital transformation so that they can be rational in their judgments. Transparency can lead to targeted involvement of employees—a decisive success factor for digital transformation—significantly reducing resistance to change (Wagner 2014). Transparency is one way to check whether a leader really stands behind what they are promoting (“walking the talk”).

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Employees must be able to witness that leaders do what they have said they will do (Kouzes and Posner 2011). A transparent leadership style encourages meaningful feedback, which allows them to make strategic adjustments that otherwise get overlooked (Tardieu et al. 2020). Rawlins (2008) and Bernstein (2017) operationalized the concept of transparency.

Working Definition: An assessment of a leader's ability to openly share information about the business strategy with employees, which is necessary for collaboration, cooperation, and collective decision-making.

As a leader of digital initiatives, I ...

1. Never hide my intentions; I reveal all aspects of our digital strategy to the organization.
2. Make sure the digital strategy is documented and accessible to all stakeholders.
3. Ensure that the benefits of our digital strategy are quantifiable so that everyone clearly understands them.

#7: Data Focus

The focus on data enables improvements in decision-making processes (Weiner, Balijepally, and Tanniru 2015), which is necessary for gaining an edge over competitors and for planning digital transformation measures (Ashwell 2017). The ability to process, analyze, prioritize, and subsequently make decisions based on data has become an essential skill for success. No seminal definition or scale exists for data focus, but it focuses on the leadership behaviors associated with business analytics and business intelligence that are necessary to transform data into business insights (Pratt and Fruhlinger 2019). Data focus is a key dimension for digital leaders because while most companies are engaged in trying to scale analytics, only a few are truly succeeding at it (Bisson et al. 2018).

Working Definition: An assessment of a leader's ability to encourage the application of data and analytics everywhere, including their own decision-making.

As a leader of digital initiatives, I ...

1. Use data analytics to drive our major operational and strategic decisions.
2. Trust data more than my intuition.
3. Make sure the right people in their organization have the training and skills necessary to work with data.

In our study, we are primarily focused on leaders' awareness of the need for effective storytelling and its use in a targeted manner.

#8: Inspire Engagement

Inspiring engagement is a central element of transformational leadership and contributes significantly to digital transformation success. Motivation and inspiration have always been essential skills that any successful leader must master (Horner-Long and Schoenberg 2002). A central aspect of this notion of inspiring engagement is the leader's ability to make employees feel they are a significant part of the company transformation and to avoid active and passive forms of resistance as much as possible. It is also about the ability to create a vision in a participatory manner—that is, staff participate in creating the vision—because transformative visions are crucial to digital transformation success. Executives must establish the connection between that company vision and the employees (Kane et al. 2019). Researchers have pointed out the lack of a generally accepted conceptualization of employee engagement or a tool to measure it (Witemeyer, Ellen, and Straub 2013). In this study, employee engagement is an attitude towards one's work in one's organization, comprising feelings of vigor, dedication, and absorption; cognitive appraisals of psychological empowerment; and motivation to act in the service of the organization's goals.

Working Definition: An assessment of a leader's ability to stimulate people's interest in or enthusiasm for doing something.

As a leader of digital initiatives, I ...

1. Let people know that their input matters and that they have the autonomy to develop and implement new ideas.
2. Create a sense of urgency, enthusiasm, and pride at work.
3. Appreciate our willingness to expend the effort necessary to persist in the face of challenges.

#9: Storytelling

Values and principles cannot make an impact on employees unless they are woven into emotional reference stories (Merath and Nau 2017). Good stories arouse emotions and visual images and thus have an entertaining and informative effect (Morgan and Dennehy 1997). In this way, the audience—whether a group or an individual—is addressed holistically both cognitively and emotionally (Lippmann, Pfister and Jörg 2019). Stories are an efficient way to disseminate information and a transformative vision; they help link the past, present, and future, create a sense of continuity among employees and make change—especially radical change—seem doable (Boal and Schultz 2007). A good story can be equally effective in informal word-of-mouth communication between employees. Therefore, organizations must empower their innovation teams to practice the art and science of innovation storytelling (Taylor 2021). Measuring storytelling primarily aims for the impact of storytelling (Rossetti and Wall 2017). In our study, we are primarily focused on leaders' awareness of the need for effective storytelling and its use in a targeted manner.

Working Definition: An assessment of the leader to inspire employees and gain buy-in.

As a leader of digital initiatives, I ...

1. Use storytelling effectively to drive the purpose of our digital transformation journey (at shareholder meetings, to staff, and to the public).
2. Continually reinforce our mission for digital transformation—my story—in daily happenings as much as possible.
3. Encourage our employees to become storytellers and ambassadors of change themselves.

#10: Digital Literacy

A deeper understanding of the technological possibilities enables leaders to create an adequate vision and inspire employees to get involved in the first place (Kane et al. 2019). If these competencies are lacking, digital leaders must develop such skills and foster them within the team to exploit the benefits of digital transformation efficiently (Cortellazzo, Bruni, and Zampieri 2019). Demonstrating relevant expertise is essential to increase a leader's credibility (Yukl 1998). Although there are scales to measure certain components of digital literacy (Jin et al. 2020; Carretero, Vuorikari, and Punie 2017), they fail to encompass the broad nature of digital literacy (Chetty et al. 2018). Since digital literacy priorities vary from industry to industry, our questions are primarily concerned with the extent to which digital leaders demand and promote digital literacy in themselves and their employees.

Working Definition: An assessment of a leader's ability to develop digital competencies within the leadership team.

As a leader of digital initiatives, I ...

1. Make sure that everyone on the direct report team has the necessary technical skills to implement our digital strategy.
2. Regularly seek outside resources (advice of thought leaders, joint research projects, partnerships, executive education, etc.) to maintain the organization's digital talents.
3. Ensure the entire leadership team knows the strategic and operational benefits of our digital technologies.

#11: Positive Attitude

A positive attitude is necessary to express emotionalizing reactions such as recognition, gratitude, and praise. Leaders with positive attitudes are more resilient, socially integrated, and high-performing, which leads to increased helpfulness, flexibility, empathy, and creativity (Creusen, Eschemann, and Johann 2010). Through positive attitudes and a positive leadership style, leaders can elevate their employees' confidence, commitment, and well-being long term (Liu, Siu, and Shi 2010). The psychometric scales in this space are more character-orientated—how this positivity manifests in the business context is more important for the digital leader.

Working Definition: An assessment of a leader's ability to carry themselves like champions and purposefully show their positivity in how they behave and communicate.

As a leader of digital initiatives, I ...

1. Advocate consistently for the best interests of our company.
2. Take it on myself to be accountable.
3. Am committed to the success of our company.

#12: Skills Acquisition

UNESCO (2018) defines digital skills as “a range of abilities to use digital devices, communication applications, and networks to access and manage information. They enable people to create and share digital content, communicate and collaborate, and solve problems for effective and creative self-fulfillment in life, learning, work, and social activities at large.” Identifying, attracting, and motivating people with these ever-evolving skill sets is of paramount importance to digital leaders. To thrive in the connected economy and society, digital skills must also function together with other abilities such as strong literacy and numeracy skills, critical and innovative thinking, complex problem solving, ability to collaborate, and socio-emotional skills. Soft skills such as agility, creativity, teamwork, and ability to learn are the skills that will survive the automated future (Maurer 2019).

Working Definition: An assessment of a leader's ability to influence digital talent acquisition and development for the organization.

As a leader of digital initiatives, I ...

1. Am personally involved in our efforts to recruit digital talents to the organization.
2. Influence efforts to retrain and rebalance the organization's digital skills.
3. Have initiated active programs with local colleges and universities to recruit digitally talented people.

#13: Knowledge Sharing

An international study of 1,700 CEOs found that outperforming organizations have leaders who create a culture of openness, connectivity, and the highest level of collaboration in the digital world (Berman and Korsten 2014). Developing a system that enables knowledge dissemination from an individual to the group is the most effective way to spread knowledge and expertise throughout an organization (Boe and Torgersen 2018). Perception- and activity-based methods are the primary approaches to measure knowledge sharing in organizations (Long, Siau, and Howell 2007). Our assessment focuses on the role of the leader, who must actively promote knowledge sharing for successful digital transformation.

Working Definition: An assessment of a leader's ability to enable access to knowledge and improve learning.

As a leader of digital initiatives, I ...

1. Insist we share our knowledge, particularly with other departments/organizations struggling with digital transformation.
2. Give every employee the opportunity to present “what they are learning/have learned” to the rest of the organization.

3. Have implemented “safety nets” that encourage experimentation and exploration of the new knowledge.

#14: Participative Style

Digital leaders are expected to strive for a leadership style in which they not only allow their team to participate but also participate in team activities themselves (Schwarz Müller et al. 2018). By doing so, they gain and maintain trust within their team (Carte, Chidambaram, and Becker 2006). Pentland (2018) showed that team-oriented leaders actively engage people in short, high-energy conversations; they are democratic with their employees’ time; they communicate with everyone equally, and they ensure that all team members get a chance to contribute. They engage in “stimulated but focused listening” and listen as much or more than they talk, and they are usually engaged with whomever they are listening to. Teaming leadership is also about creating open spaces, giving team members freedom to choose their efforts, and motivating them to voluntarily make a positive effort to complete tasks (Rahmani, Roels, and Karmarkar 2018). Servant leadership emphasizes “increased service to others; a holistic approach to work; promoting a sense of community; and the sharing of power in decision making” (Spears, Greenleaf, and Covey 2002, p. 4). Several measurement scales—for example, the one proposed Bhatti et al. (2019)—have examined the effects of participative, team-oriented leadership and organizational citizenship behavior relationships.

Working Definition: An assessment of a leader’s ability to develop a collaborative and inclusive way to approach problems and opportunities.

As a leader of digital initiatives, I ...

1. Require that we celebrate “small wins,” as well as major accomplishments, to motivate employees.
2. Prefer “rolling up my sleeves” and working directly with the development teams.
3. Make time for people who have questions.

#15: Track Record

This dimension reflects a transactional facet of digital leadership, but regarding the leader themselves. This critical

reflection enables a down-to-earth self-assessment—perhaps the leader needs to further build competencies, deal with new disruption, or improve digital literacy. For track record, successful transformation is also always about measurable results, whether positive or negative (Prince 2018). Several business results metrics could be included here—for example, improved outcomes and increased customer experience. Rather than listing them out, we chose to keep the measurement at a higher level—namely, did the digital initiative produce meaningful results?

Working Definition: An assessment of the leader’s digital initiatives to produce meaningful results.

As a leader of digital initiatives, I ...

1. Changed an aspect of business using digital methods.
2. Reduced the cycle time or operational cost of the business significantly.
3. Created an overall better customer experience.

Managerial Implications

We hypothesized and found support for the fact that the human dimensions of digital leaders are the foundation of digital transformation success. We recommend leaders focus on these 15 key dimensions we outline in our Digital Leadership Scale. We highlight three key takeaways:

1. Building trust and credibility are at the core of effective digital leadership. Credible leaders walk the talk, and employees learn by observing their leaders.
2. Employees should be at the center of your digital strategy. Align employees with business goals and clearly communicate your plan and vision for the company to them. Listen to your employees, understand what they are thinking, and integrate employee feedback into your communications and operations strategy. Doing so will help you build an innovative, transparent, and inclusive culture.
3. Digital leaders must make conscious efforts to ensure trustworthiness in the use of digital technologies such as AI and machine learning. They must ensure these technologies are used in a way that’s fair, respectful of individual privacy, and beneficial to society.

Future Research

We identified 15 human dimensions of digital leadership that we consider key to digital transformation success. Given the urgency and importance of digital transformation and the critical role digital leaders play in successful execution, we are currently working on two initiatives.

We are developing a Digital Leadership Scale to measure these 15 dimensions on a five-point Likert scale. We will administer this scale to digital leaders. Using statistical methods (for example, principal component analysis), we will develop a statistically valid and reliable scale that measures key dimensions of digital leadership.

Based on the scale, we plan to create a user-friendly self-assessment tool and make it available for free to anyone leading a digital initiative—at the project, function, business

Participatory leaders engage in “stimulated but focused listening” and listen as much or more than they talk, and they are usually engaged with whomever they are listening to.

unit, or company level. We emphasize the term self-assessment because we intend for the scale to inspire leaders to reflect and improve upon aspects of their leadership based on science rather than others' expectations of them.

Limitations

The dimensions presented should be checked for their distinctiveness. Within the research discussions, for example, it emerged that the Storytelling dimension can be a sub-item of the Inspire Engagement dimension. Measuring character traits is admittedly difficult, especially since the measurements are mostly self-reported questionnaires. As a reflexive assessment, such a measurement fulfills the criterion of practical relevance and can be classified as useful. Nevertheless, we recommend a supplementary review and discussion of the dimensions to increase objectivity and usefulness. Additionally, the inductive reasoning approach used is limited as it begins with our observation of the role of digital leadership in digital transformation success. Our observations are based on a set of interviews, yet we may have inadvertently overlooked important factors. Future research should focus on quantitative studies to investigate the aspects found in this study—namely, its intercorrelations and their relevance depending on industry and company demographics.

Conclusion

We presented a prescriptive assessment for leaders in digital transformation by identifying the human dimensions required to successfully lead a digital transformation initiative. In line with the motto “reflection enables insight for improvement,” we intend for the Digital Leadership Scale to inspire leaders to improve their leadership style based on science. Leaders can use the 15 dimensions we identified for critical self-reflection. The overarching insight derived from this study is that the human dimensions of leadership are critical to a successful digital transformation.

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